

Shopping, Demand Composition and Equilibrium Prices

Presented by Nicolò Ceneri - BGGMP Reading Group, NYU

Lukas Nord

Job Market Paper

European University Institute

Motivation

- ▶ Posted prices for identical products display significant dispersion
 - ▶ Households do not pay the same prices **for identical products**
(Aguiar and Hurst 2007) Price Differences
 - ▶ Households **do not buy the same products** along the expenditure distribution
(Argente and Li 2021, Handbury 2021) Basket Overlap
- Price distribution depends on the distribution of households

QUESTION

How does the distribution of households affect markups and inequality?

The Missing Link

- ▶ Incomplete markets
(Bewley 1977, Aiyagari 1994, Huggett 1993,...)
- ▶ Search for consumption
(Burdett and Judd 1983, Kaplan and Menzio 2016,...)
- ▶ Incomplete markets + Search for consumption **but** one variety
(Arslan et al. 2021, Pytka 2022)

THIS PAPER:

- ▶ Incomplete markets + Search for consumption + Non-homotheticities

Today's Agenda

Evidence

- Shopping effort

- Demand Composition

Model

- Households' problem

- Retailer's problem

- Model Fit

Implications

- Expenditure Inequality

- Average Prices and Markups

Evidence - Overview

- ▶ **Data:** Nielsen Consumer Panel
- ▶ How does shopping behavior change across households?
 - Heterogeneity in prices paid by households
 - Number of stores visited
- ▶ What is the relationship between shopping and price distribution?
 - Exploit variation in demand composition across products

Do richer households spend more for identical items?

► Price index:

- Construct households' price index [Details](#)

$$P_{it} = \left(\frac{X_{it}}{\tilde{X}_{it}} - 1 \right) \times 100$$

- Regress HH's price index on household FE η_i , year-state FE α_{st} , controls Z_{it}

$$P_{it} = \eta_i + \alpha_{st} + \gamma Z_{it} + \epsilon_{it}$$

► Trips per purchase:

- i 's shopping trips normalized by the number of purchases

Table 1: Shopping Effort across Households

	price index (1)	trips per purchase (2)
log(expenditure)	0.706*** (0.078)	-0.042*** (0.001)
income 30k-60k	0.080* (0.046)	-0.001* (0.001)
income 60k-100k	0.178*** (0.057)	-0.002** (0.001)
income >100k	0.326*** (0.070)	-0.002** (0.001)
1 non-employed household head	-0.236*** (0.037)	0.002*** (0.000)
2 non-employed household heads	-0.422*** (0.068)	0.004*** (0.001)
mean		0.15
FE year-state	X	X
FE household	X	X
Observations	801,398	801,398

Note: Regression of shopping effort on household characteristics. Column (1) index of individual prices paid vs. national annual average price. Column (2) annual number of shopping trips divided by number of purchases. Data from Nielsen Consumer Panel waves 2007-2019. Observations weighted with Nielsen provided sample weights. Standard errors clustered at the household level. *p<0.1; **p<0.05; ***p<0.01.

Does shopping effort affect retailers' margins?

► First best:

Direct measure of markups at the seller-good level.

- Not observed in the data

► Second best:

Demand-weighted shopping effort vs skeweness of posted prices Skewness

- High demand-weighted search → tougher competition within varieties

$$\text{skew}_{j,r,t} = \underbrace{\theta_m}_{\text{FE product module}} + \underbrace{\mu_{r,t}}_{\text{FE time-region}} + \underbrace{\sum_{g=2}^G \beta_g \text{share}_{j,g,y(t)}}_{\text{Expenditure share of HH group for variety}} + \varepsilon_{j,r,t}$$

Table 2: Demand Composition and the Skewness of Price Distributions

	by expenditures			by income		by employment
	all	working age		working age		all
	(1)	(2)		(3)		(4)
expenditure quintile 2	-1.638*** (0.242)	-1.467*** (0.206)	income 30k-60k	-0.136 (0.133)	1 non-employed household head	0.864*** (0.115)
expenditure quintile 3	-2.309*** (0.256)	-2.076*** (0.221)	income 60k-100k	-0.824*** (0.155)	2 non-employed household heads	1.011*** (0.210)
expenditure quintile 4	-3.067*** (0.258)	-2.582*** (0.219)	income >100k	-0.820*** (0.139)		
expenditure quintile 5	-3.412*** (0.253)	-3.007*** (0.224)				
FE product module	X	X		X		X
FE quarter-SMA	X	X		X		X
Observations	3,026,551	3,026,404		3,026,404		3,026,551

Note: Regression of the skewness of price distributions on demand shares by household groups. Price distributions defined as all transactions of a barcode within a Scantrack Market Region and quarter. Demand shares defined as the share of national annual spending on a barcode by each group of households. Data from Nielsen Consumer Panel waves 2007-2019. Observations weighted by total sales in given price distribution. Standard errors clustered at the barcode-year level. *p<0.1; **p<0.05; ***p<0.01.

- ▶ Continuum of infinitely lived households with
 - Heterogeneous labor earnings zw
 - **Non-homothetic preferences** over varieties j
 - **Incomplete information** of posted prices $\rightarrow$ Search for second retailer
- ▶ Measure one of retailers with homogeneous cost κ_j within varieties

Model - Households' problem

- First stage (Divide cash-on-hand over expenditure and savings):

$$\begin{aligned} V(z, a) &= \max_{e, a' \geq 0} U(e) + \beta \mathbb{E}_{z'|z} V(z', a') \\ \text{s.t.} \quad e + a' &\leq (1 + r)a + zw \end{aligned}$$

- Second stage (Consumption among varieties and **shopping effort**):

$$\begin{aligned} U(e) &= \max_{s \in [0, 1], \{c_j\}_{j=1}^J, c_O} u(\mathcal{C}) - v(s, \mathcal{C}) \\ \text{s.t.} \quad \mathcal{C} &= (c_G)^\alpha (c_O)^{1-\alpha} \\ c_G &= \left[\sum_{j=1}^J (\mathcal{C})^{\frac{q_j}{\sigma}} (c_j)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \\ c_O + \sum_{j=1}^J p_j(s) c_j &\leq e. \end{aligned}$$

Households' Optimality Conditions - Shopping Effort

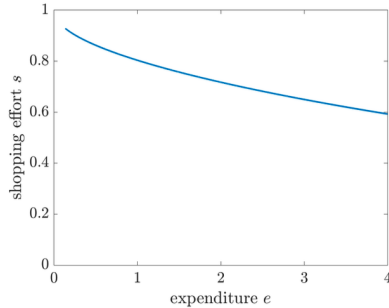
Deriving the FOC wrt s

$$\underbrace{-v_s(s, \mathcal{C})}_{\text{marginal disutility of shopping}} = \underbrace{\frac{\partial p_G(\mathcal{C}, s)}{\partial s} c_G}_{\text{change in BC}} \underbrace{(1 - \alpha) \frac{\mathcal{C}}{c_O} [u'(\mathcal{C}) - v_C(s, \mathcal{C})]}_{\text{marginal benefit of relaxing BC}}$$

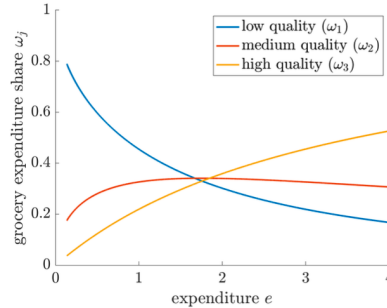
We get a relationship between price index and shopping effort

$$\frac{\partial p_G(\mathcal{C}, s)}{\partial s} = p_G(\mathcal{C}, s)^\sigma \sum_{j=1}^J \mathcal{C}^{q_j} (p_j(s))^{-\sigma} \underbrace{\frac{\partial p_j(s)}{\partial s}}_{\text{return to shopping}}$$

The Effect of Expenditure: Shopping Effort vs Expenditure shares



(a) Shopping Policy



(b) Grocery Expenditure Shares

Figure 3: Shopping Effort and Consumption Baskets

Note: Model implied shopping effort s and expenditure shares ω_j across grocery varieties j as a function of total household expenditure e . Expenditure shares are computed as total spending of a household on variety j divided by her total grocery expenditures $e^G = \alpha e$.

Model - Retailers' problem

Retailers post prices **taking into account** $\bar{s}_j = \int \frac{\lambda_j c_j(x_i)}{C_j} s_j(x_i) di$

$$\tilde{\pi}_j(p) = \underbrace{\frac{C_j}{M_j}}_{\text{market size}} \underbrace{\left[(1 - \bar{s}_j) + \bar{s}_j 2(1 - F_j(p)) \right]}_{\text{market share}} \underbrace{(p - \kappa_j)}_{\text{margin}} - \underbrace{K_j}_{\text{Fixed cost}}$$

In equilibrium **non-degenerate distribution** (Burdett and Judd 1983, Pytka 2022)

$$\underbrace{C_j [(1 - \bar{s}_j) + \bar{s}_j 2(1 - F_j(p))]}_{\text{current sales}} = \underbrace{C_j [\bar{s}_j 2 f_j(p)]}_{\text{loss in sales}} \underbrace{(p - \kappa_j)}_{\text{current margin}}$$

Model - Equilibrium

A **stationary equilibrium** consists of:

- ▶ $V(z, a), \{C_O(z, a), \{c_j(z, a)\}_{j=1}^J\}, s(z, a), e(z, a), a'(z, a)$
- ▶ $\lambda(z, a), \{C_j\}_{j=1}^J, \{\bar{s}_j\}_{j=1}^J, \{F_j(p)\}_{j=1}^J, \{p_j(s)\}_{j=1}^J$

such that

- ▶ Given $\{p_j(s)\}_{j=1}^J$ households choose optimally
- ▶ Given $\{\bar{s}_j\}_{j=1}^J$, the posted price distribution solves the retailers' problem
- ▶ The distribution of households λ is stationary
- ▶ $C_j = \iint \lambda(z, a) c_j(z, a) dz da. \quad \bar{s}_j = \iint \frac{\lambda(z, a) c_j(z, a)}{C_j} s(z, a) dz da.$

- ▶ Calibrate income process

(De Nardi et al. 2020)

- ▶ Set functional form and exogenous parameters

Exogenous Parameters

- ▶ Target wealth-to-income ratio, price dispersion and expenditure composition

Targeted Moments

Validation - Demand Composition

Table 5: Demand Composition of Grocery Varieties

	quintile of expenditures				
	Q1	Q2	Q3	Q4	Q5
low quality (q_1)	0.1138	0.1752	0.2107	0.2360	0.2642
medium quality (q_2)	0.0555	0.1228	0.1838	0.2521	0.3859
high quality (q_3)	0.0249	0.0770	0.1432	0.2405	0.5145

Note: Model implied demand shares by varieties of the grocery good and expenditure quintile. Demand shares are computed as total sales of variety j to quintile g divided by total sales of variety j .

Validation - Price Distribution

Table 6: Price Distributions of Grocery Varieties

		quality of grocery variety		
		low (q_1)	medium (q_2)	high (q_3)
demand-weighted shopping effort	$\bar{s}_j$	0.76	0.73	0.71
average price posted	μ_j^F	1.49	1.60	1.83
average price paid	μ_j^G	1.38	1.48	1.71
standard deviation of posted prices	σ_j^F	0.30	0.31	0.33
return to search	$\frac{\partial p_j(s^i)}{\partial s^i}$	-0.15	-0.16	-0.17
skewness of posted prices	γ_j^F	1.51	1.42	1.33

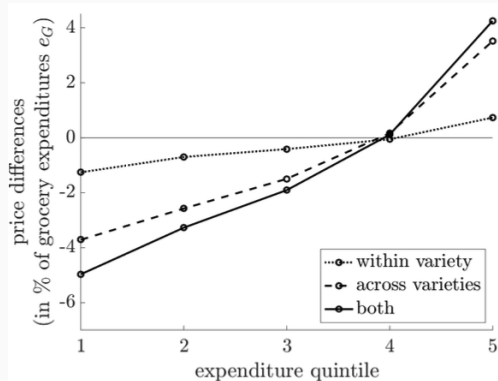
Note: Summary statistics of model implied price distributions for each grocery variety j .

Table 7: Income Distribution – Model vs. Data

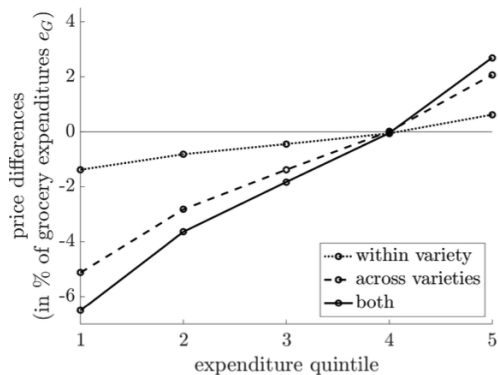
	quintile of post-tax income				
	Q1	Q2	Q3	Q4	Q5
Model	5.02%	10.50%	15.78%	23.89%	44.79%
Data	6.44%	10.91%	14.70%	20.31%	47.65%

Note: Fit of the model implied income distribution. In the model, income is measured as labor and financial income ($zw + ra$). Data moments for household income after taxes and transfers from Congressional Budget Office (CBO) for 2007-2018.

Price Differences vs Expenditure Distribution

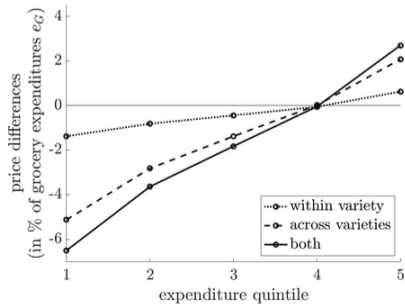


(a) Data

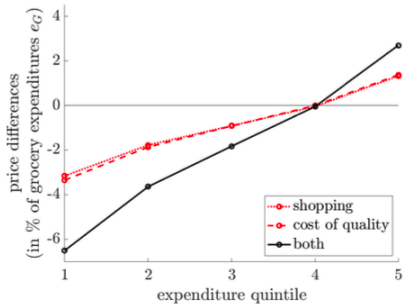


(b) Model

Figure 5: Price Differences along the Expenditure Distribution



(a) Model (Original)



(b) Model (Adjusted)

Figure 6: Price Differences along the Expenditure Distribution - The Effect of Shopping

Note: Price differences by expenditure quintile in the model, as a share of households' grocery spending. In the *original* decomposition, “within variety” refers to differences between the price paid by a given household and the average price for a given product (direct effect of shopping), while “across varieties” is the difference in average prices across different products. In the *adjusted* decomposition, “shopping” refers to price differences within and margin differences across products (direct and equilibrium effect of shopping) and “cost of quality” to differences in marginal costs across products.

Markups Implications: Change in Demand Composition

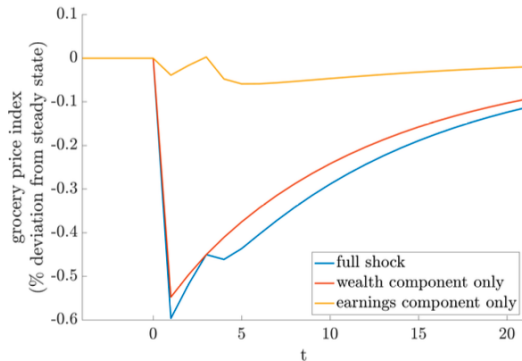


Figure 11: Response of Posted Prices to Losses in Earnings and Wealth

Note: Model implied response of an aggregate Laspeyres index $P_t^F = \frac{\sum_{j=1}^J C_j^{SS} \mu_{jt}^F}{\sum_{j=1}^J C_j^{SS}}$ of prices posted to the Great Recession shock (15% loss in wealth and earnings losses from Heathcote et al. (2020)). Full shock decomposed into response to loss in earnings and loss in wealth.

Conclusions

THIS PAPER

- ▶ Augmented previous theories of households heterogeneity with search frictions

HOW

- ▶ Non-homothetic preferences
- ▶ Retailers' optimally choose given demand-weighted search effort

SO WHAT?

- ▶ Wedge of about 5% between consumption and expenditure inequality
- ▶ Heterogeneous response to demand shocks across varieties

THANK YOU!

Construction of the Price Index

- ▶ Total cost of HH i 's consumption bundle

$$X_{it} = \sum_j p_{jit} c_{jit}$$

- ▶ Counterfactual with quantity-weighted **average price across all HH**

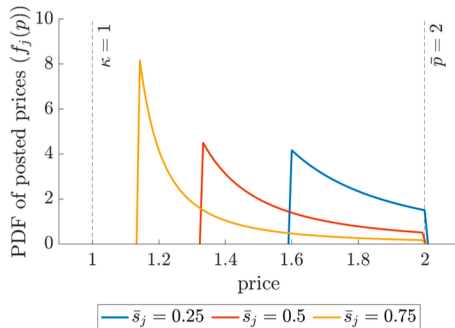
$$\tilde{X}_{it} = \sum_j \tilde{p}_{jt} c_{jit}$$

Price index: ratio between true and counterfactual cost

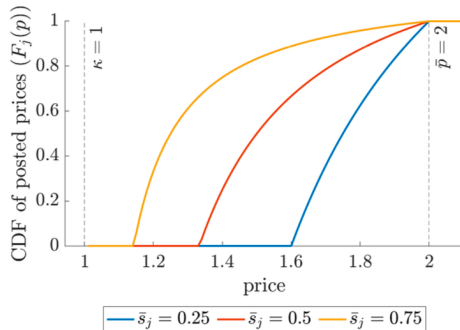
$$P_{it} = \left(\frac{X_{it}}{\tilde{X}_{it}} - 1 \right) \times 100$$

Difference in the cost of i 's consumption bundle relative to paying average prices

Skewness of Price Distribution



(a) Density



(b) Cumulative Density

Figure 1: Posted Price Distributions

Note: Price distributions derived from retailers' optimal price posting problem for marginal cost $\kappa_j = 1$, maximum willingness to pay $\bar{p}_j = 2$, and three levels of demand-weighted shopping effort $\bar{s}_j$.

Calibration Targets and Model Fit

Table 3: Calibration Targets and Model Fit

Target	Data	Model	Source
basket overlap (Q1 vs. Q5)	63.28%	63.76%	Nielsen (2007-2019)
Δp across varieties (Q1 vs. Q5)	7.2%	7.2%	Nielsen (2007-2019)
Δp within varieties (Q1 vs. Q5)	2%	2%	Nielsen (2007-2019)
$\text{mean}(CoV_j)$	0.1920	0.1915	Nielsen (2007-2019)
$CoV_2 - CoV_1$	-0.0120	-0.0078	Nielsen (2007-2019)
$CoV_3 - CoV_1$	-0.0203	-0.0211	Nielsen (2007-2019)
average markup	1.39	1.39	ARTS (2007-2019)
wealth/income	3	3	

Note: Results of the internal calibration of $(\psi_1, \psi_2, \kappa_2, \kappa_3, \bar{p}_1, \bar{p}_2, \bar{p}_3, \beta, q_1, q_3)$.

Calibration Parameters

Table 4: Calibrated Parameter Values

Parameter	Value	Notes
J	3	
σ	2.0	Jaravel and Olivi (2021)
α	0.35	Broda and Parker (2014)
ϕ	2.0	
r	0.02	
q_j	$[-0.67 \ 0 \ 0.67]$	
ψ_1	0.0023	
ψ_2	0.21	
κ_j	$[1 \ 1.06 \ 1.219]$	
$\bar{p}_j$	$[2.55 \ 2.65 \ 2.90]$	$2.55 + 1.6(\kappa_j - \kappa_1)$
β	0.9232	

Note: Summary of calibrated parameter values. $(J, \alpha, \sigma, \phi, r)$ are set externally. $(\psi_1, \psi_2, \kappa_2, \kappa_3, \bar{p}_1, \bar{p}_2, \bar{p}_3, \beta, q_1, q_3)$ are calibrated internally.

Optimal allocation of consumption

- ▶ Grocery consumption is a non-homothetic consumption aggregator
- ▶ The demand system can be characterized through expenditure shares

$$\frac{\omega_j}{\omega_k} = \mathcal{C}^{q_j - q_k} \left(\frac{p_j(s)}{p_k(s)} \right)^{1-\alpha}$$

- ▶ q_j represents the expenditure elasticity of demand
- ▶ If $q_j > q_k$ then the share of variety j vs variety k ($\frac{\omega_j}{\omega_k}$) $\uparrow$ expenditure e

Externalities among households

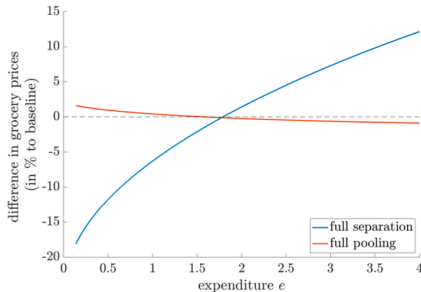


Figure 9: Shopping Externality

Note: Change in households' prices paid under alternative posted price distributions. Baseline is the calibrated steady-state. Counterfactuals fix the expenditure distribution, consumption baskets and shopping effort. *Full pooling* assumes price distributions for each variety determined as the best response to average search effort $\bar{s}_j = \int \int \frac{\lambda(z,a) \sum_{j=1}^J c_j(z,a)}{\sum_{j=1}^J C_j} s(z,a) dz da$. *Full separation* assumes individually targeted price distributions for each variety determined as the best response to individual search effort $\bar{s}_j = s(z,a)$.

Decompose Price Responses

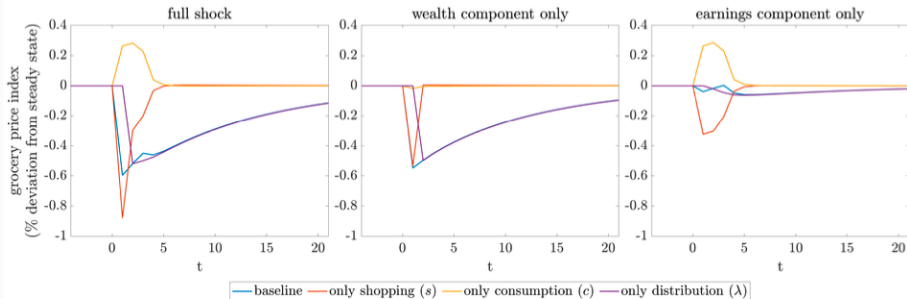


Figure 12: Decomposition of Posted Price Responses

Note: Response of an aggregate Laspeyres index of posted prices $P_t^F = \frac{\sum_{j=1}^J C_j^{SS} \mu_{jt}^F}{\sum_{j=1}^J C_j^{SS}}$ to the Great Recession shock (15% loss in wealth and earnings losses from Heathcote et al. (2020)). Panels report responses to the full shock, only the wealth, and only the earnings component. Complete response as a baseline, for each panel decomposed into the response to changes in consumption policies, shopping policies, and the distribution of households, holding the respective others constant at steady state levels.

Basket Overlaps Across Distribution

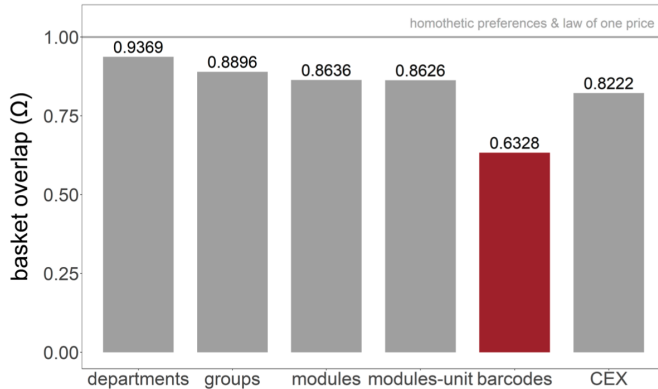


Figure 15: Consumption Basket Overlap - Top vs. Bottom Expenditure Quintile

Note: Histogram overlap in the vector of expenditure shares for the bottom and top quintile of expenditures, by different definitions of a product. First five columns derived from Nielsen Consumer Panel, final column from Consumer Expenditure Survey (CEX). CEX column considers 14 spending categories.

Price Differences Across Distribution

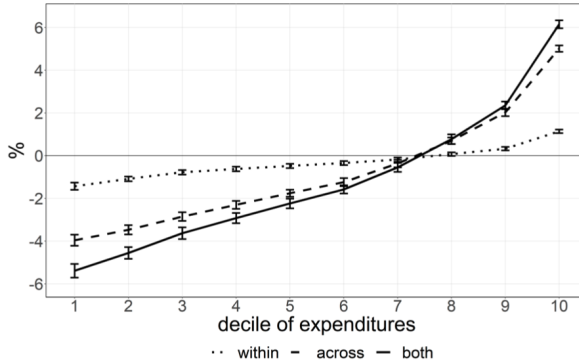


Figure 17: Price Differences along the Expenditure Distribution – Deciles

Note: Price differences by expenditure quintile in the data, as a share of households' grocery spending. "Within" refers to differences between the price paid by a given household and the average price for a given product (direct effect of shopping), while "across" is the difference in average prices across different products. Data from the Nielsen Consumer Panel 2007-2019. The price within products is computed by barcode and the price across products is computed across barcodes within a product module (by unit of measurement). Confidence intervals bootstrapped with 100 draws at the household-year level.

Price Differences along the Expenditure Distribution

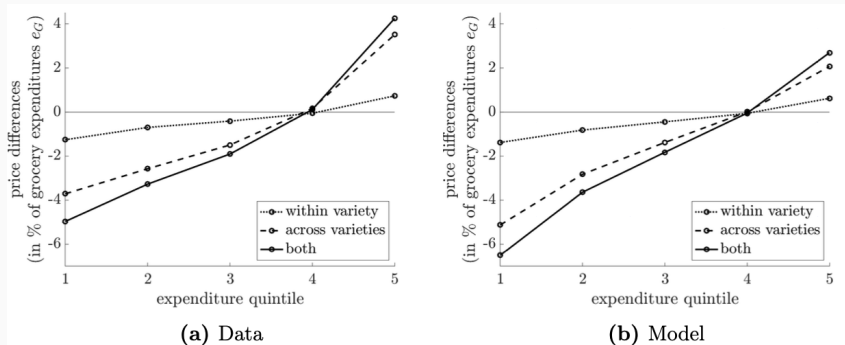


Figure 5: Price Differences along the Expenditure Distribution

Consumption Inequality

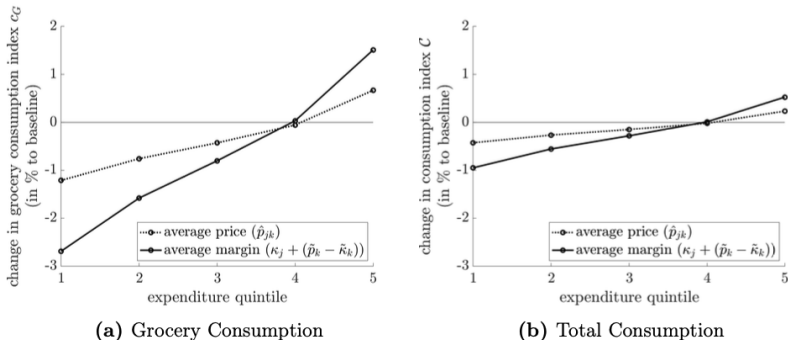


Figure 7: Shopping Effort and Consumption Inequality

Note: Change in households' grocery (c_G) and total consumption ($\mathcal{C}$) absent shopping effort. Counterfactuals fix the expenditure distribution and let households choose a basket assuming they have to pay (i) the average price within each grocery variety (shutting down the direct effect of shopping) or (ii) marginal cost plus the average margin across varieties (shutting down the direct and equilibrium effect).

Decompose Effect of Shopping

$$\begin{aligned}
 e_i^G &= \underbrace{\sum_k \sum_{j \in J_k} (p_{ijk} - \hat{p}_{jk}) c_{ijk}}_{\text{(direct effect of shopping)}} + \underbrace{\sum_k \sum_{j \in J_k} (\hat{p}_{jk} - \tilde{p}_k) c_{ijk}}_{\text{across varieties}} + \sum_k \sum_{j \in J_k} \tilde{p}_k c_{ijk} \\
 &= \underbrace{\sum_k \sum_{j \in J_k} (p_{ijk} - \hat{p}_{jk}) c_{ijk} + ((\hat{p}_{jk} - \kappa_{jk}) - (\tilde{p}_k - \tilde{\kappa}_k)) c_{ijk}}_{\text{shopping (direct+equilibrium)}} \\
 &\quad + \underbrace{\sum_k \sum_{j \in J_k} (\kappa_{jk} - \tilde{\kappa}_k) c_{ijk}}_{\text{cost of quality}} + \sum_k \sum_{j \in J_k} \tilde{p}_k c_{ijk}
 \end{aligned}$$

Go Back